

1-B:

Fixed costs do not change with output in the short run, so as production increases these costs are spread across more units. Since average fixed cost equals total fixed cost divided by output, the figure must fall continuously as output rises. This is unlike other cost measures, which typically fall and then rise. Hence, key B is correct.

Why the others are wrong:

A: Average cost combines fixed and variable costs. It falls due to economies of scale, but then rises due to diseconomies of scale.

C: Average variable cost also falls initially but rises once diminishing returns occur, so it does not fall continuously.

D: Marginal cost declines at first but later increases as Marginal Productivity of Labor falls, so it does not steadily decrease.

2-D:

Market failure arises when private market decisions do not achieve the socially efficient allocation of resources. Merit goods such as education or healthcare are under-consumed because individuals underestimate their benefits. The market equilibrium lies below the socially optimal output, creating under-provision that governments often correct through subsidies or direct provision.

Why the others are wrong:

A: A firm selling below Average cost and making a loss represents predatory or below cost pricing. **B:** Accumulation of unsold stocks from a guaranteed minimum price is a distortion caused by government intervention, not a failure of the free market itself.

C: Low-income earners unable to afford goods reflects inequity rather than market failure.

3-D:

A cartel needs to restrict output and keep prices above competitive levels. If entry is easy due to **lower barriers to entry**, abnormal profits attract new competitors who increase supply and weaken the cartel's ability to maintain high prices. Effective collusion is far more difficult when outsiders can enter freely. Hence, key D becomes correct.

A: A high concentration ratio means **that** a small number of firms dominate an industry, suggesting limited competition, making collusion easier.

B: Limit pricing deters potential entrants, helping cartel members keep control.

C: High fixed costs raise barriers to entry, supporting collusion rather than undermining it.

4-B:

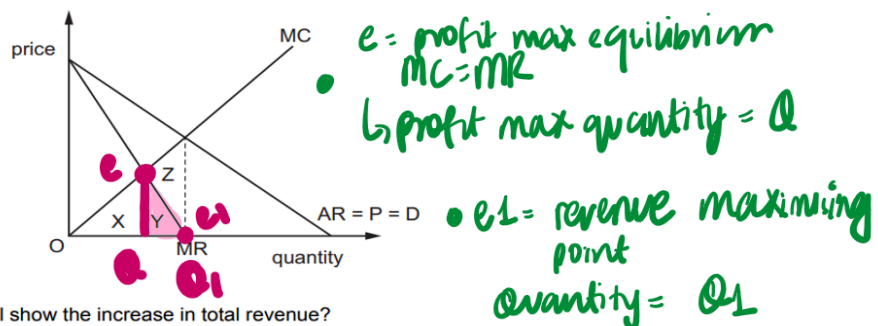
Marginal revenue is the additional revenue a firm earns from selling one more unit of output. Marginal revenue is calculated by dividing the change in total revenue by the change in total output.

This means that the change in total revenue between two output levels equals the sum of marginal revenue over that quantity range. On a diagram, this change is represented by the **area under the MR curve** between the initial and final quantities.

A profit-maximising firm produces where $MR = MC$, but a revenue-maximising firm produces where $MR = 0$, because total revenue reaches its peak when its slope is zero. Moving from the profit-maximising output to the revenue-maximising output increases total revenue by exactly the area **Y** under the MR curve between these two quantities.

This area measures the cumulative additional revenue from selling the extra units.

The diagram shows a firm's cost and revenue curves. The firm changes its objective from profit maximising to revenue maximisation.



Which area on the diagram will show the increase in total revenue?

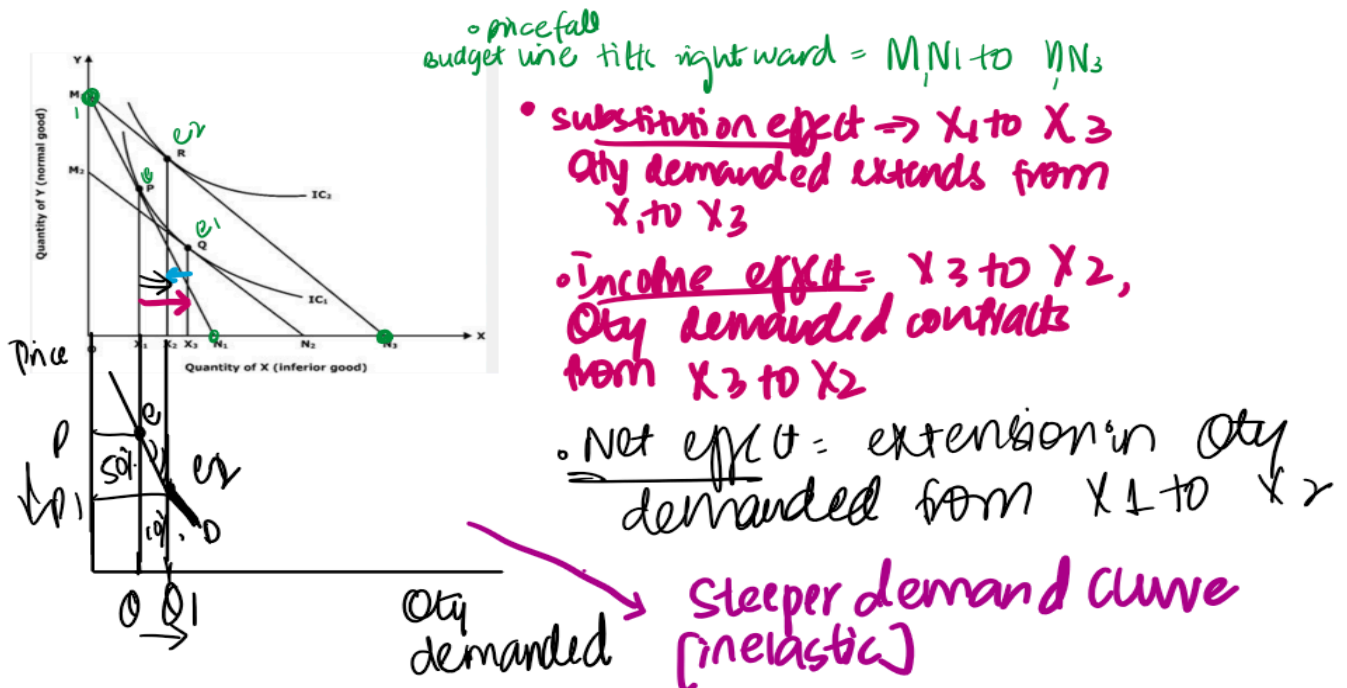
- A $X + Y + Z$ **B** Y C $Y + Z$ D Z

5-A:

For an inferior good, a price fall encourages consumers to substitute toward it, raising quantity demanded. However, the higher real income from the price fall reduces demand because consumers prefer superior goods. **These effects oppose each other.** Demand still slopes downward overall, but the net change in quantity is smaller, creating a relatively steep/ inelastic curve. Hence, key A is correct.

B and D: These suggest income and substitution effects act in the same direction, which is true for normal goods, not inferior goods.

C: The income effect for an inferior good is negative rather than positive.

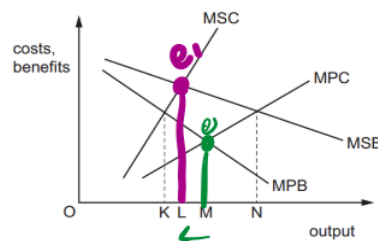


6-C;

The free market produces at M, where marginal private costs equal marginal private benefits representing “free market equilibrium”, but marginal social costs and marginal social benefits are equal at L, representing the “socially optimal level of output”

Decreasing output from M to L aligns production with the socially efficient level, reducing deadweight loss and internalising the externality. **This requires output to fall to where marginal social cost equals marginal social benefit. Hence, making key C correct.**

6 The diagram shows the private and social costs and benefits of production in a free market that result in market failure.



Which change in output would be necessary to overcome this market failure?

- A from K to M
- B from M to N
- C from M to L**
- D from N to L

socially equilibrium
level of output = L
($MSC = MSB$)

private output = M.
($MPC = MPB$)

7-B:

Productive efficiency improves, dynamic efficiency improves, allocative efficiency unchanged.

Using profits to invest in better capital equipment reduces production costs, improving productive efficiency. (fall in ATC is associated with improving productive efficiency)

Innovation over time reflects dynamic efficiency as new technology is adopted. (The ATC curve at the same level of output will shift downwards)

However, monopoly pricing still sets price above marginal cost, so allocative efficiency does not improve unless regulation forces prices to fall.

8-D:

Subsidising a merit good increases output toward the social optimum level. This is an example of successful intervention, not failure. Government failure occurs when policy causes inefficiency or unintended consequences worse than the original problem.

Why the others are wrong:

A: A price ceiling set too low discourages supply, driving firms out of the market.

B: A guaranteed minimum price set too high causes overproduction, resulting in a surplus in the market, which can get wasted.

C: A high minimum wage can lead to unemployment when firms dismiss workers.

9-A:

Progressive taxation places a higher burden on high earners relative to low earners, reducing post-tax income differences and reducing inequality.

This is because progressive taxes include direct taxes, such as income taxes, which take a greater percentage of the incomes of the rich than the poor. This policy directly improves vertical equity and lowers measures such as the Gini coefficient. Hence, key A becomes correct.

Why the others are wrong:

B: Making someone better off without harming others (Pareto improvement) says nothing about distribution.

C: When the best use is made of scarce resources, this represents efficiency (this does not show anything about inequality)

D: An increase in the Gini coefficient indicates rising inequality, not greater equality.

10-B:

A: This is a valid reason for imposing a tax based on pollution. When firms are taxed according to the amount they pollute, the cost of pollution becomes internalised into their production decisions. By reducing emissions, firms lower their tax burden, so the policy encourages cleaner production methods.

B: This is *not* a reason to impose such a tax, it is a disadvantage. Pollution taxes require accurate measurement of emissions and regular monitoring of firms, which can be costly and technically difficult. This makes option B correct.

C: This supports the use of a pollution-based tax because it is fair. Firms that pollute less will pay less tax, while heavy polluters pay more. This helps align private costs with social costs and ensures that firms face charges in proportion to the harm they create.

D: This is also a valid reason. A pollution tax increases producers' costs, reducing output and discouraging harmful practices, thereby moving production closer to the socially efficient quantity.

11-B:

A. A specific tax charges a fixed amount per unit, while an ad valorem tax is imposed as a percentage. This doesn't explain why the tax revenue was lower than forecast.

B. This is the most likely reason. If fewer than originally forecast number of drinks contained sugar, this would reduce the tax revenue collected by the government (as indirect tax is a tax imposed on goods sold)

C. Whether or not prices rose, the tax would still be collected per unit, so this doesn't explain the shortfall.

D. Inelastic demand means consumers keep buying despite price increases after imposition of tax. If true, tax revenue should have met or exceeded forecasts. This contradicts the outcome and cannot explain the lower revenue.

12-A:

A real wage is the money/nominal wage adjusted for changes in inflation. If nominal pay rises by less than the rate of inflation, purchasing power falls and real wages decline. By holding pay awards below inflation, the real cost of labour is reduced, meeting the stated aim. Hence, making key A correct.

Why the others are wrong:

B: Linking private pay to company profits does not guarantee real wages will fall, as profits may rise faster than inflation.

C: Introducing a minimum wage generally raises pay for low earners, lifting rather than reducing real wages.

D: Removing restrictions on unions strengthens wage bargaining power and is likely to push nominal wages up, risking higher real wages rather than lower.

13-A:

The marginal revenue product of labour measures the additional revenue generated by employing one more unit of labour.

Since $MRP = MPP \times MR$, it combines the productivity of the worker (marginal physical product) with the revenue gained per extra unit of output (marginal revenue).

14-A:

Economic rent is the difference between actual earnings and transfer earnings (the minimum needed to keep a worker in a job). Here, her transfer earnings as a teacher are \$25 000. Since she actually earns \$30 000, her economic rent is $\$30\,000 - \$25\,000 = \$5\,000$.

Why the others are wrong:

B: We don't know her transfer earnings if she *were* an accountant (what she'd need to stay in that job). The question gives no reservation wage for accountancy, so this figure of economic rent can't be inferred and is not supported.

C: Transfer earnings as a teacher are \$25 000, not \$30 000.

D: No information suggests \$15 000 as the minimum to keep her in accountancy; this is unsupported and therefore incorrect. The figure provided is simply her potential accounting salary.

15-C:

A) Changes in consumption leading to changes in income represents the multiplier. The formula of multiplier is: $\text{Change in RNI} / \text{Change in Aggregate expenditure}$.

B) Changes in income leading to changes in consumption represents marginal propensity to consume; the formula of MPC is: $\text{Changes in consumption} / \text{Changes in income}$.

C) How changes in income lead to changes in investment represents the **accelerator principle**. The accelerator principle states that investment spending depends on the rate of change of income. When national income rises quickly, firms invest more to expand productive capacity because they expect future demand to remain high. When income growth slows, investment falls sharply.

The **accelerator coefficient** is given by:

$$\beta = \frac{\Delta I}{\Delta Y}$$

Where:

- β = accelerator coefficient
- ΔI = change in investment
- ΔY = change in income (or output)

D) Changes in investment leading to changes in income represents the multiplier as well.

16-C:

A production possibility curve shows the maximum combinations of goods that can be produced with existing resources and technology.

A move from a point inside the curve i.e. the actual (X) to a point on the curve i.e. the potential (Y) means previously idle resources are now fully employed. This is called **actual growth** because the economy moves closer to its productive potential without shifting the curve itself.

A: A positive output gap occurs when output exceeds the long-run potential, which is not shown here.

B: A recession would move the economy closer to the origin (actual moving closer to the origin, instead of moving closer to the potential)

D: Potential economic growth is represented by an outward shift of the PPC due to more resources or better technology.

17-A:

Quantity Theory of Money

The Quantity Theory of Money is expressed by the equation: $MV=PT$

Where:

- M = Money supply
- V = Velocity of circulation (how quickly money changes hands)
- P = Price level
- T = Number of transactions (or real output)

What Velocity Means?

The velocity of money, V , is the **average number of times each unit of money is used for transactions in a given period**. It measures how fast money circulates in the economy/ or how fast money changes hands in an economy.

A. A move to monthly rather than weekly salary payments by firms means that workers are paid less frequently. This means that they must keep money unused or hold it for longer periods to cover expenses between paydays. This means that on average, money circulates more slowly **because it sits idle for longer**. As a result, the income velocity of circulation, V , falls. This is the best explanation.

B. Higher interest rates **make holding idle cash unattractive because it earns no return**. People respond by moving money quickly into interest-bearing accounts rather than letting it sit in cash form. This behaviour causes each unit of money to circulate more rapidly, which raises V rather than reducing it.

C. If people can withdraw cash whenever they need it, they do not have to store large sums at home or in their wallets. Money is drawn in smaller amounts and spent quickly, with balances turning over more often. **Since money is not kept idle in hand**, this also raises V rather than lowering it.

D. When credit cards are used less, people need to hold larger cash balances to pay for goods and services, which **'may'** reduce V . However, this effect is weaker and less direct as it depends on a lot of external factors, such as income levels and consumer confidence, as compared to option A, making A the better answer.

18-C:

Occupational mobility is the ease with which workers can switch between different jobs or industries. Training programmes provide workers with new skills that match labour market requirements, enabling them to move into alternative occupations more readily. This reduces structural unemployment by making workers more flexible.

A: Lower income tax raises disposable incomes of workers, which can boost their consumption levels but does not equip them with new skills.

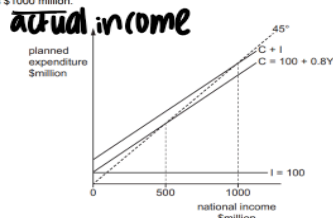
B: Reducing unemployment benefits might encourage job search thereby reducing voluntary unemployment but does not increase the ability to change occupations.

D: More advertising of vacancies improves information flow thereby reducing frictional unemployment but cannot overcome skill mismatches.

19-A:

Step 1: Identification of the positive output gap, and differentiating it from the inflationary gap can be seen below:

19 The diagram shows a closed economy with no government. It is in initial equilibrium when the national income is \$1000 million.



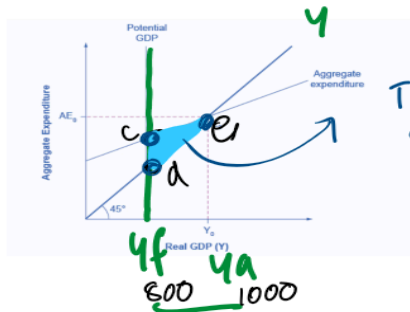
positive output gap

Actual income - potential income
 $1000 \text{ mn} - 800 \text{ mn}$
 $\Rightarrow 200 \text{ mn}$

This is the positive output gap and not the inflationary gap

* full employment (potential income) = \$800 mn
 If the full employment national income occurs at \$800 million, what is the value of the inflationary gap?

- A \$40 million B \$160 million C \$260 million D \$840 million



This is the inflationary gap

4a > 4f: positive output gap of "200"
 +ve output gap

Then we need to find out the MPC to calculate the multiplier, and then find out the inflationary gap:

$C = 100 + 0.8Y$
 $C = a + bY_d \rightarrow \text{disposable income}$
 autonomous consumption MPC

Step 2: Use the MPC to find multiplier

multiplier = $\frac{1}{1 - \text{MPC}} \Rightarrow \frac{1}{1 - 0.8} \Rightarrow \frac{1}{0.2} \Rightarrow 5$

Step 3:

inflationary gap $\times$ multiplier = positive output gap
 $40 \times 5 = 200 \rightarrow \frac{200}{5} \Rightarrow 40$

Multiplier $k = 1/(1-0.8)=5$. A gap in income of 200 (1000–800) corresponds to an initial spending excess of $200/5 = 40$. Therefore the inflationary gap in spending terms is **\$40 million**.

Hence, key A, 40 mn is correct.

20-A:

Internal value of money

- The internal value of money refers to its **real value or its purchasing power at home** i.e. how much domestic goods and services can be bought with one unit of currency.
- If prices in country X rise, each unit of currency buys **fewer goods domestically**, so the internal value of money **falls**.

External value of money

- The external value of money refers to the exchange rate i.e. how much foreign currency one unit of domestic currency can buy.
- If prices rise only in country X while staying constant in country Y, goods in X become less competitive internationally. Foreign buyers purchase fewer of X's goods, so demand for X's currency falls, because foreigners need less of it to pay for those goods.
- At the same time, imports into X become relatively cheaper. Residents of X buy more goods from abroad, which requires them to sell X's currency and buy foreign currency, increasing the supply of X's currency on foreign exchange markets.
- Falling demand plus rising supply of X's currency pushes its exchange rate down, reducing the external value of money.
- Therefore, the external value of money also falls.

21-A:

Subsidies to producers reduce firms' production costs, shifting aggregate supply to the right. As a result, there is non-inflationary growth, i.e. income increases alongside a simultaneous reduction in the general price level of an economy. Unlike demand-side measures, which risk overheating when resources are near full employment, supply-side subsidies improve both output and price stability. Therefore, key A becomes correct.

Why the others are wrong:

B: Increasing government spending is an expansion of aggregate demand, which raises output but may also cause inflation if the economy is already near capacity.

C: Lowering direct taxes increases disposable income and consumption, again stimulating

Aggregate demand, which can trigger demand-pull inflation.

D: Lower interest rates promote borrowing and spending, which increases Aggregate demand, raising demand-pull inflation if spare capacity is limited.

22-D:

Buying foreign machinery increases imports of capital equipment. In the short run, this **raises the value of imports and can push the current account into deficit**. Hence, key D becomes correct.

Why the others are wrong:

A: A fall in tax revenue is a fiscal issue, which affects the budget, not directly a current account factor.

B: Export-led growth would tend to increase exports and improve, not worsen, the current account.

C: Debt repayments affect the financial account rather than the current account, since they are capital flows.

23-A:

A rising price level makes a country's goods less competitive abroad, reducing foreign demand for its exports and therefore for its currency. At the same time, domestic consumers buy more imports, increasing supply of the currency on foreign exchange markets. Both effects cause depreciation, not appreciation. Hence, making key A correct.

B: Higher domestic interest rates attract foreign capital inflows, creating demand for the currency and strengthening it.

C: Higher incomes abroad raise foreign demand for exports, increasing demand for the domestic currency, which will result in appreciation of the country.

D: More tourists visiting the country buy the domestic currency to spend locally, supporting appreciation.

24-B:

Borrowing from the non-bank private sector to finance deficit → money supply unchanged

1) When the government sells bonds to households or firms (the non-bank sector), people pay using their existing deposits. The government then **spends that same money** back into the economy. The deposits that left private accounts to buy the bonds **return to private accounts when the government spends**.

Net result: no new money is created, the money supply stays the same — it just shifts from private savers to government recipients.

Here are the reasons for all the options:

A: Borrowing from the non-bank private sector does not increase money supply or lower interest rates, so there is no automatic pressure on the currency to fall. This is not correct.

B: The increase in real output will be limited as the value of the multiplier is low. With an MPC of 0.2, the multiplier is only 1.25. Even though deficit spending raises demand, the effect is small. This is why this option is correct.

C: When the government sells bonds to people and firms, they pay with existing deposits. The government then spends this money back into the economy, so the total money supply stays the same.

D: Borrowing from the non-bank sector does not inject new money (money supply remains unchanged), so interest rates do not automatically fall.

25-D:

Option D is correct because it accurately reflects the components of the balance of payments. The current account does include trade in services and secondary income. The financial account does include official reserve assets, which are managed by the central bank, and also portfolio investments.

Option A is incorrect because it places official reserve assets in the current account. Official reserve assets are part of the financial account, not the current account. While trade in goods is correctly classified under the current account, the inclusion of reserve assets in the same category is a misclassification.

Option B is incorrect because it misplaces secondary income in the financial account. Secondary income, such as remittances and foreign aid, is a component of the current account. Although official reserve assets are correctly placed in the financial account, the incorrect classification of secondary income makes this option invalid.

Option C is incorrect because it includes secondary income in the financial account. Secondary income belongs in the current account. While foreign direct investment is correctly placed in the financial account, the misplacement of secondary income makes this incorrect. Trade in goods and services is correctly listed under the current account, but the error in the financial account components makes this option incorrect.

26-B:

The graph shows GDP per head on one axis and inequality (Gini coefficient) on the other. A country with the lowest GDP per head is typically least developed because it has low average income. Since country X has the lowest gdp per head, it is the least developed, hence making key B correct.

Option A, We can get an idea of Average incomes using the Real GDP/head data. Since real gdp/head of country Z is higher than X, we can imply that Z has higher average incomes.

Option C, Income is not equally distributed in country Y, as it has the highest value of gini coefficient relative to other countries, implying that income is unequally distributed.

Option D, The distribution of income in country Y is less even than in country X, as Y has higher gini coefficient than X.

27-C:

This scenario (increase in birth rate and a fall in the death rate) describes demographic changes that lead to an increase in the total population size. **Without corresponding improvements in productivity, technology, or resource availability, a larger population may lead to diminishing returns and a lower per capita income.** This is why, key C, becomes correct.

All the other options are directly improving quality/quantity of factors of production. An increase in optimum population is typically associated with factors that enhance productive capacity or resource availability, allowing a larger population to be sustained at a higher standard of living.

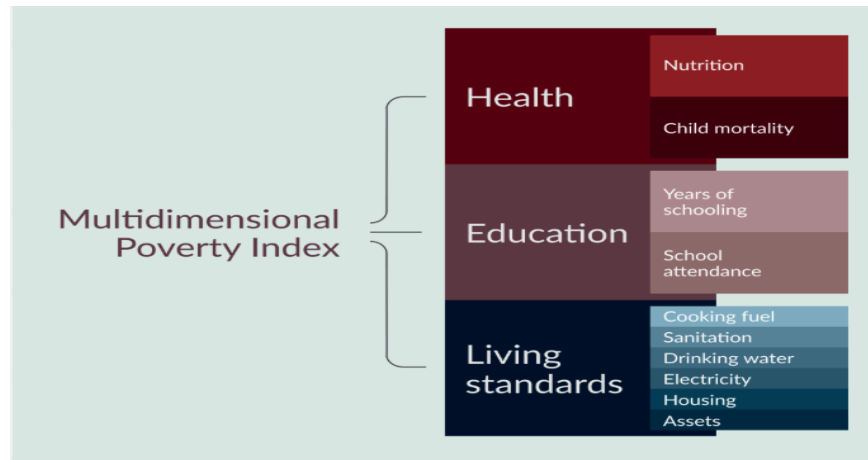
Option A, Improvement in Labour Productivity: When labour productivity improves, a given amount of labour can produce more output, leading to a higher per capita income and potentially supporting a larger optimum population. This is represented by an increase in output per worker.

Option B, Improvement in the Quality of Capital and Technology: Advancements in capital and technology increase the efficiency of production, allowing for greater output from the same or fewer inputs. This can lead to a higher optimum population as more people can be supported at a higher standard of living.

Option D, Increase in the Availability of Natural Resources: An increase in the availability of natural resources, such as land, minerals, or energy, expands the productive capacity of an economy, enabling it to support a larger population at an optimal level.

28-C:

While health is considered via child mortality and nutrition, **specific access to medication** is not an individual weighted component of the MPI, as can be seen below in the image outlining the components of MPI.



Why the others are wrong: A, B, D: Cooking fuel, electricity, and water are core dimensions of living standards included in the MPI.

29-D:

The World Bank focuses on long-term development projects such as roads, dams, and schools. It provides loans and grants to finance infrastructure. The IMF, by contrast, concentrates on short-term balance of payments support and macroeconomic stabilisation. Hence, key D is correct.

A: Facilitating free trade is a role of the WTO rather than either the IMF or World Bank.

B: The IMF provides policy advice on macroeconomic management, including fiscal and monetary policy frameworks, particularly for countries facing balance of payments issues.

C: Bailouts for countries with external account crises are mainly the IMF's responsibility.

30-C:

This suggests that foreign aid has created dependency rather than self-sustaining growth. When local economies rely heavily on outside expertise and funding, they may fail to build their own capacity, thereby leading to long term development being undermined. Therefore, key C becomes correct.

A: Improved capacity to benefit from technology transfer is evidence that aid is effective, not ineffective.

B: Investment in infrastructure showing new transport links is a positive result of aid.

D: Education and training projects that enhance the supply side demonstrate successful development outcomes.

